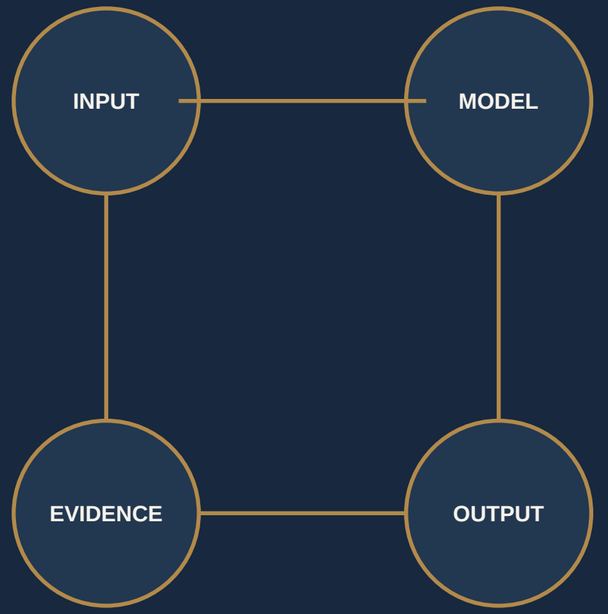
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Mobile Robotics

AI.207 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define wheeled robots.
- Apply key methods including differential drive.
- Build or analyze a practical system involving omnidirectional systems.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Wheeled robots

02

METHODS

Differential drive

03

SYSTEMS

Omnidirectional systems

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Wheeled robots

Wheeled robots

02 · Differential drive

Differential drive

03 · Omnidirectional systems

Omnidirectional systems

04 · Locomotion

Locomotion

05 · Sensors

Sensors

06 · Localization

Localization

07 · Navigation and mobile-robot c

Navigation and mobile-robot control

08 · Mandatory practical: Simulate

Mandatory practical: Simulate and program a mobile robot to navigate an environment

Core ideas, made explicit

AI.207 • Distinguish definitions, assumptions and evidence

- Wheeled robots
- Differential drive
- Omnidirectional systems
- Locomotion
- Sensors

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.207 • Distinguish definitions, assumptions and evidence

- Localization
- Navigation and mobile-robot control
- Mandatory practical: Simulate and program a mobile robot to navigate an environment

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Wheeled robots

INPUTS
Differential drive

METHOD
Omnidirectional systems

OUTPUT
Locomotion

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving wheeled robots.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using wheeled robots.
- Implement or simulate a baseline using differential drive.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.